Topic: Fundamental solution sets and the Wronskian

Question: Find the Wronskian of the fundamental set of solutions.

$$\{e^{-2x}, e^{3x}\}$$

Answer choices:

A $W(e^{-2x}, e^{3x}) = -5e^{-x}$

B $W(e^{-2x}, e^{3x}) = -5e^{x}$

C $W(e^{-2x}, e^{3x}) = 5e^{x}$

D $W(e^{-2x}, e^{3x}) = 5e^{2x}$

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Solution: C

Given the solution set $\{y_1, y_2\} = \{e^{-2x}, e^{3x}\}$, the Wronskian will be

$$W(y_1, y_2) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = y_1 y_2' - y_2 y_1'$$

$$W(e^{-2x}, e^{3x}) = \begin{vmatrix} e^{-2x} & e^{3x} \\ -2e^{-2x} & 3e^{3x} \end{vmatrix} = (e^{-2x})(3e^{3x}) - (e^{3x})(-2e^{-2x})$$

Simplifying the determinant, we get

$$W(e^{-2x}, e^{3x}) = 3e^{x} + 2e^{x}$$

$$W(e^{-2x}, e^{3x}) = 5e^{x}$$

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Topic: Fundamental solution sets and the Wronskian

Question: Which solution set cannot be a fundamental set of solutions?

Answer choices:

A $\{e^{3x}, 7e^{-x}\}$

B $\{2e^{-x}, 7e^{x}\}$

C $\{e^{-x}, 7e^{-x}\}$

D $\{3e^{-x}, 7e^{x}\}$

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Solution: C

A solution set will only be a fundamental set of solutions when the Wronskian of the set is non-zero. So the set that doesn't form a fundamental set will be the set for which $W(y_1, y_2) = 0$.

For answer choice A, the Wronskian is

$$W(e^{3x},7e^{-x}) = \begin{vmatrix} e^{3x} & 7e^{-x} \\ 3e^{3x} & -7e^{-x} \end{vmatrix}$$

$$W(e^{3x},7e^{-x}) = (e^{3x})(-7e^{-x}) - (7e^{-x})(3e^{3x})$$

$$W(e^{3x},7e^{-x}) = -7e^{2x} - 21e^{2x}$$

$$W(e^{3x},7e^{-x}) = -28e^{2x}$$

$$W(e^{3x},7e^{-x}) \neq 0$$

For answer choice B, the Wronskian is

$$W(2e^{-x},7e^{x}) = \begin{vmatrix} 2e^{-x} & 7e^{x} \\ -2e^{-x} & 7e^{x} \end{vmatrix}$$

$$W(2e^{-x},7e^{x}) = (2e^{-x})(7e^{x}) - (7e^{x})(-2e^{-x})$$

$$W(2e^{-x},7e^{x}) = 14 + 14$$

$$W(2e^{-x},7e^{x}) = 28$$

$$W(2e^{-x},7e^{x}) \neq 0$$

For answer choice C, the Wronskian is

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$$W(e^{-x},7e^{-x}) = \begin{vmatrix} e^{-x} & 7e^{-x} \\ -e^{-x} & -7e^{-x} \end{vmatrix}$$

$$W(e^{-x},7e^{-x}) = (e^{-x})(-7e^{-x}) - (7e^{-x})(-e^{-x})$$

$$W(e^{-x},7e^{-x}) = -7e^{-2x} + 7e^{-2x}$$

$$W(e^{-x},7e^{-x}) = 0$$

For answer choice D, the Wronskian is

$$W(3e^{-x},7e^{x}) = \begin{vmatrix} 3e^{-x} & 7e^{x} \\ -3e^{-x} & 7e^{x} \end{vmatrix}$$

$$W(3e^{-x},7e^{x}) = (3e^{-x})(7e^{x}) - (7e^{x})(-3e^{-x})$$

$$W(3e^{-x},7e^{x}) = 21e^{-x}e^{x} + 21e^{x}e^{-x}$$

$$W(3e^{-x},7e^{x}) = 21 + 21$$

$$W(3e^{-x},7e^{x}) = 42$$

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Topic: Fundamental solution sets and the Wronskian

Question: Find the Wronskian of the fundamental set of solutions of the differential equation, generated with $y(0) = 1$ and $y'(0) = 0$, and $y(0) = 0$ and $y'(0) = 1$.

$$y'' + y = 0$$

Answer choices:

A 1

B e^x

C $\cos(2x)$

D $\cos x - \sin x$

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Solution: A

Normally, we would find the general solution to this second order equation by solving the associated characteristic equation,

$$r^2 + 1 = 0$$

$$r = \pm i$$

and then plugging these roots into the formula for the complementary solution with complex conjugate roots.

$$y_c(x) = c_1 \sin x + c_2 \cos x$$

The derivative of this complementary solution is

$$y_c'(x) = c_1 \cos x - c_2 \sin x$$

If we now plug in the initial conditions $y(x_0) = 1$ and $y'(x_0) = 0$, and $y(x_0) = 0$ and $y'(x_0) = 1$, we'll be able to generate another fundamental set of solutions. We'll use $x_0 = 0$, since we can choose any value of x_0 that we like, and using $x_0 = 0$ will be easiest.

For $y(0) = 1$, $\quad 1 = c_1 \sin 0 + c_2 \cos 0$

For $y'(0) = 0$, $\quad 0 = c_1 \cos 0 - c_2 \sin 0$

Then this system gives $c_1 = 0$ and $c_2 = 1$, so we could say that one solution in our fundamental set of solutions is

$$y_1(x) = \cos x$$

The second set of initial conditions gives

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For $y(0) = 0$, $0 = c_1 \sin 0 + c_2 \cos 0$

For $y'(0) = 1$, $1 = c_1 \cos 0 - c_2 \sin 0$

Then this system gives $c_1 = 1$ and $c_2 = 0$, so we could say that the other solution in our fundamental set is

$$y_2(x) = \sin x$$

So one fundamental set of solutions is $\{y_1, y_2\} = \{\cos x, \sin x\}$, but another is

$$\{y_1', y_2'\} = \{-\sin x, \cos x\}$$

Now we can find the Wronskian.

$$W(y_1, y_2) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$$

$$W(y_1, y_2) = \begin{vmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{vmatrix}$$

$$W(\cos x, \sin x) = \cos^2 x + \sin^2 x$$

$$W(\cos x, \sin x) = 1$$

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